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each answer is submitted to the correct question

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Your work may be submitted on your own paper, by writing on this worksheet printed out, or by writing via tablet, among possibly other options. If you are having problems submitting your work, please attend office hours. (There is no late penalty for submitting the assignment on the due date, but there will be no late submissions accepted for any reason.) Click on [Gradescope's help page](#) and read their instructions.

1. Compute the following antiderivative $\int \frac{1}{x^2 - 1} dx$

2. Compute the following antiderivative: $\int \frac{\sec^2 \theta}{\tan^2 \theta - 1} d\theta$

Hint: As a first step, use a substitution to convert this into an integral which requires partial fractions.

3. Compute the following antiderivative: $\int \sec \theta \tan^2 \theta \, d\theta$

4. Using the previous problem, compute the following antiderivative: $\int \sqrt{x^2 + 2x} \, dx$

5. When using a numerical method to approximate the value of an integral, we expect there to be some discrepancy between the exact value and the value found by our computation. We cannot (usually) determine exactly what this error is, but it is possible to show that the (absolute value of the) error in using Simpson's rule to approximate $\int_a^b f(x) dx$ cannot exceed the bound

$$E(n) = \frac{(b-a)^5}{180n^4} M$$

where M is the maximum value of $|f''''(x)|$ on the interval $[a, b]$.

- (a) Compute $\int_0^{\pi/2} \sin(x) dx$.
- (b) Compute the Simpson's rule estimate of this integral using $n = 4$ to at least 4 decimal places.
- (c) State the error in using this approximation; i.e., state the difference between the exact value in (a) and the approximation in (b) to at least 4 decimal places.
- (d) Find the value of M for this problem. That is, find the maximum value of $|f''''(x)|$ on the interval $[0, \frac{\pi}{2}]$ where $f(x) = \sin(x)$. You may do this by sketching a graph of $|f''''(x)|$ and then determining what the maximum is, or by using calculus techniques.
- (e) Find the error bound $E(n)$ for approximating $\int_0^{\pi/2} \sin(x) dx$ using Simpson's rule with $n = 4$, $n = 10$, and $n = 20$. That is, find $E(4)$, $E(10)$, and $E(20)$ to at least 4 decimal places.
- (f) Find the first (valid) value of n which is large enough that the error bound $E(n)$ is smaller than 0.0001.
- (g) Find the first (valid) value of n which is large enough that the error bound $E(n)$ is smaller than $10^{-9} = 0.000000001$.

6. Refer to the previous problem.

(a) Compute $\int_1^4 \frac{1}{x} dx$.

(b) Compute the Simpson's rule estimate of this integral using $n = 4$ to at least 4 decimal places.

(c) State the error in using this approximation; i.e., state the difference between the exact value in (a) and the approximation in (b) to at least 4 decimal places.

(d) Find the value of M for this problem. That is, find the maximum value of $|f''''(x)|$ on the interval $[1, 4]$ where $f(x) = \frac{1}{x}$. You may do this by sketching a graph of $|f''''(x)|$ and determine what the maximum is, or by using calculus techniques.

(e) Find the error bound $E(n)$ for approximating $\int_1^4 \frac{1}{x} dx$ using Simpson's rule with $n = 4$, $n = 10$, and $n = 20$. That is, find $E(4)$, $E(10)$, and $E(20)$ to at least 4 decimal places.

(f) Find the first (valid) value of n which is large enough that the error bound $E(n)$ is smaller than $10^{-9} = 0.000000001$.

7. Suppose you need to integrate some specific function f . From experimental data, you know that $f''(x) = e^{-x^2}$ and that

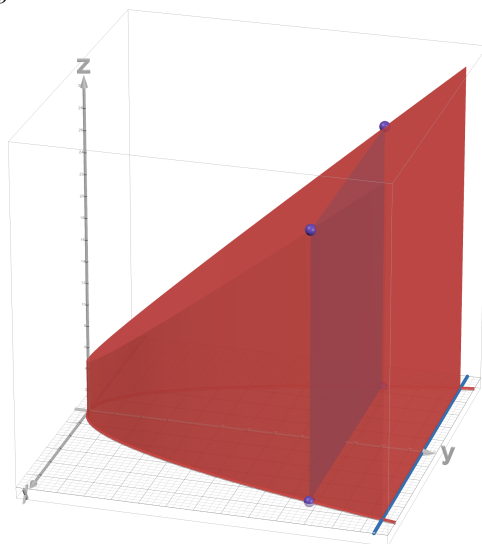
x	-0.5	-0.25	0	0.25	0.5
$f(x)$	7.870	7.906	8.000	8.156	8.370

- (a) Calculate the trapezoidal approximation T_4 for the integral $\int_{-0.5}^{0.5} f(x) dx$ to 3 decimal places.

- (b) Find the maximum that the error might be, $E(4)$, for the above approximation to at least 3 decimal places. The error formula for the trapezoidal approximation T_n looks like $E(n) = \frac{(b-a)^3}{12n^2} M$ where M is the maximum value of $|f''(x)|$ on the interval $[a, b]$.

- (c) Combining the previous two parts, determine an interval which is guaranteed to contain the true value of the integral. Provide at least 3 decimal places for the endpoints.

8. Suppose a new theater being designed has a base which can be described as the region bounded by $y = x^4$ and $y = 25$.



The cross-sections perpendicular to the y -axis will be rectangles whose base is 4 units shorter than its height.

- (a) Find an exact value A so that the line $y = A$ divides the floor into two equal areas.
- (b) Find the (approximate) percentage of the volume of the theater which is in the region $y \geq A$. Give your answer to two decimal places.